

Topic: Projection Operation

Q. Discuss the Projection operation in Relational Algebra. Describe its syntax and explain its behavior regarding duplicate elimination, using suitable examples.

Definition to Remember

The **Projection** operation is used to select specific columns (attributes) from a relation, resulting in a new relation with only those columns.

Projection Operation (π)

* Operates on a **single table** (unary).

* Filter

2. Syntax of Projection

$\pi_{\{ \text{attribute_list} \}}(\text{Relation_Name})$

* **attribute_list**: Comma-separated column names to retain.

3. Examples

Input Relation: STUDENT

| Roll_No | Name | Major | City |

| :--- | :-

Example 1: Simple Projection

Query:

Example 2: Duplicate Elimination

Query:

Example 3: Composition - Selection + Projection

Query:

*****Must-Write Points (for 10 marks)*****

1. Proje

*****Quick Recall*****

`pi (Pi)
sigma`